

Aaron Besnoff

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EDUCATION

Lehigh University, Bethlehem, PA

Graduation: Spring 2027

Bachelor of Science in Mechanical Engineering; Minor in Aerospace Engineering

Advanced Coursework: Advanced Engineering Dynamics, Control Systems, Vibrations, Aircraft Design, Astrodynamics, Aerospace Engineering, Manufacturing, Fluid Mechanics, Thermodynamics

INDUSTRY EXPERIENCE

Engineering Intern – CEVA Logistics

June 2025 – August 2025

- Proposed and developed a **macro automation solution** that saved the company **500+ labor hours** in label creation and implementation.
- Assisted in the **planning, development, and implementation** of a new Warehouse Management System (WMS).
- Designed a **Pick Pathing optimization system** to reduce inbound-to-outbound transfer times and improve workflow efficiency.

RESEARCH & DESIGN EXPERIENCE

ONR MURI – Lab Research

January 2024 – August 2025

Lehigh University, Moored Laboratory, Bethlehem, PA

- Contributed to the **ONR MURI** Non-Traditional Hydrodynamic Propulsion program through active involvement in the project "Bio-Inspired Flexible Propulsors for Fast, Efficient Swimming: What Physics are we Missing?"
- Assisted on the execution of hydrodynamic experiments to gather and analyze data for various research projects.
- Utilized SolidWorks to improve on previously created experimental designs.
- Machined and assembled equipment for the "Pitch and Heave" hydrofoil project.

Bio-Inspired Multi-Medium UAV (Air/Water) – Lab Research

August 2025 – Present

Lehigh University, Moored Laboratory, Bethlehem, PA

- Collaborating with another undergraduate student to develop a **bio-inspired flapping-wing drone** capable of sustained operation in both aerial and underwater environments.
- Currently designing and prototyping a **waterproof thermal test stand** informed by principles of **fluid dynamics, biomimicry, and aerospace design** to evaluate water resistance and heat dissipation capabilities.
- Utilizing iterative design and testing to address challenges in **waterproof sealing and propulsion efficiency** across different mediums.

Bomb Squad EOD X-Ray Deployment Robot – Drivetrain and Controls

January 2026 – Present

Lehigh University Senior Capstone

- Leading drivetrain and electromechanical system integration for a low-cost teleoperated EOD robot, designing a four wheel independently driven platform around portable X-ray payload, terrain, packaging, and cost requirements.
- Architecting a Raspberry Pi/Python control system linking an ELRS handheld radio to four RS485 hub motors, including skid-steer driving, closed loop speed/current control, operator-adjustable torque modes, and handling of sequential motor feedback and communication latency.
- Developed the 18 V lithium-ion power architecture including motor and battery selection, BMS integration, charging, power distribution, wiring, and protection, as well as defining failsafe behavior for radio loss, motor faults, emergency stopping, and deliberate system re-arming.

Supersonic Aircraft Design Project

January 2025 – May 2025

- Designed and analyzed a supersonic air-superiority aircraft using **NASA OpenVSP/VSPAERO**, including aerodynamic configuration, sizing, and mission-performance analysis.

Founder - Lehigh University Rocketry Association

January 2024 – August 2025

- Worked on initial design for a liquid rocket motor thrust chamber; designed and built L1/L2 rockets using SolidWorks, OpenRocket, 3D printing, and machining.

TECHNICAL SKILLS

- Software/Programming:** SolidWorks, Python, Abaqus FEA, OpenVSP/VSPAERO, ANSYS STK, Mastercam
- Development Tools:** Github, Claude Code, Codex
- Manufacturing:** CNC Milling, Lathe, Waterjet, 3D printing, Injection Molding